

Mohammed Ashraf Ahmed Elwardany

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OBJECTIVE

Results-driven Full-Stack Engineer (**Node.js, Next.js, PostgreSQL**) with experience building mission-critical systems at the Egyptian Space Agency. Proficient in **AI-assisted development & Vibe Coding** architecture. Adept at transforming complex requirements into secure, scalable, and high-performance applications.

TECHNICAL SKILLS

Frontend : React.js, Next.js, React Native, Redux/Context API, Tailwind CSS.
Backend : Node.js, Express.js, REST APIs.
Databases & ORM : PostgreSQL, MongoDB, Prisma.
Tools & Architecture : Git/Gitee, Figma (UI/UX Implementation), i18n (Localization).
AI & Productivity : Vibe Coding workflows, AI-assisted development (Cursor/Copilot).

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer
Egyptian Space Agency (EgSA)
Software Engineering & Architecture:

- **Orbit Propagation System:** Created and deployed an API and web application using Node.js, React.js, and MongoDB.
- **Mobile Development:** Created a cross-platform React Native application to deliver real-time orbit session schedules for the NEXSAT mission.
- **Front-End Portals:** Designed dynamic, high-performance web platforms (Agriculture & EODD Portals) using React.js, implementing advanced state management and custom theming (Dark/Light modes) resulting in a 30% improvement in load times and seamlessly serving over 5,000 daily users.

International Collaboration & Localization:

- **LASAC Cloud Platform:** Represented EgSA in China to architect cloud platform localization. Successfully implemented complex i18n multi-language support (English/Arabic), including RTL styling and custom font configurations.

Operations & Data Management:

- **NEXSAT Operations:** Assumed full operational responsibility for the software and communication pipelines of Egypt's first TRL-9 satellite.
- **Data Processing:** Processed and managed vital satellite image data for the MisrSat-2 project.

03/2024 – Present
Cairo, Egypt

INTERNSHIPS

Embedded Software Engineering Intern
National Telecommunication Institute (NTI)
Strengthened low-level system architecture and memory management concepts, directly contributing to writing highly optimized backend code.

09/2023 – 11/2023

Space Apps Intern
EgSA - Egyptian Space Agency

07/2022 – 09/2022

IoT Application Development Intern
Information Technology Institute (ITI)
Focused on connecting hardware sensors to cloud architectures and web interfaces via APIs.

02/2022 – 03/2022

Mobile Cross-Platform Intern
Information Technology Institute (ITI)
Gained hands-on experience in cross-platform frontend development, extending full-stack capabilities to mobile environments.

08/2021 – 09/2021

EDUCATION

Bachelor of Science in Computer Science

Faculty of Computers and Information, Suez University

06/2023
Suez, Egypt

- **Grade:** Very Good (GPA: 3.08 / 4.0)
- **Graduation Project:** Space Rover - Real-Time Mission Control Web System (Grade: A+)
- **Full-Stack Web Development:** Developed a comprehensive, real-time mission control dashboard. Built the interactive frontend using **React.js** and engineered a high-performance **.NET** backend server.
- **Real-Time Data Streaming:** Implemented **WebSockets** to establish low-latency, bidirectional communication between the web server and edge devices.
- **Hardware-to-Web Integration:** Successfully integrated the web platform with an ESP32-based autonomous rover, enabling web-based control of motors, camera feeds, and sensor arrays.

PROJECTS

Full-Stack Engineer | PM Academy

11/2024 – present

E-Learning & PMP Exam Simulation

- **Full-Stack Architecture:** Architected and deployed an end-to-end e-learning and exam simulation platform using **Next.js, Node.js, and PostgreSQL**.
- **Backend & Database:** Designed scalable relational database schemas and developed secure RESTful APIs to handle complex business logic, user progress tracking, and exam evaluations.
- **Frontend Integration:** Built a dynamic web interface to ensure seamless user experience and fast load times.

Full-Stack Developer | Quantum Middle East Corporate Website & CRM

Client Location: UAE

End-to-End Development: Led the full-stack development of a modern corporate portal tailored for the UAE market.

System Integration: Bridged a responsive, state-managed **React.js/Redux** frontend with a robust **Flask (Python)** backend.

Custom CMS/CRM: Designed and implemented a custom backend dashboard to streamline dynamic content management and client data handling.

CERTIFICATIONS & AWARDS

3rd Place Winner (Team Award)

Space Apps Hackathon,
Egyptian Space Agency (EgSA).

Software Engineering Summer Training

Egyptian Space Agency (EgSA).

Mobile Cross-Platform
Information Technology
Institute (ITI).

Summer trainig - Suez Steel

Introduction to IoT
Certification

Information Technology
Institute (ITI).

ATFEL English Certificate
(score 500)